MIL-B-12128C(MR)
13 August 1965
SUPERSEDING
MIL-B-12128B(Ord)
10 January 1962

MILITARY SPECIFICATION

BRASS, SINTERED; STRUCTURAL PARTS

1. SCOPE

1.1 Scope. This specification covers brass sinterings of three compositions.

1.2 Classification.

1.2.1 Composition. The sinterings shall be furnished in the following compositions, as specified (see 3.1):

Composition 1 - 90/10 copper-zinc alloy
Composition 3 - High leaded 80/20 copper-zinc alloy
Composition 5 - 70/30 copper-zinc alloy

1.2.2 Classes. The sinterings shall be furnished in the following classes, as specified (see 3.2):

Class A
Class B

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal from a part of this specification to the extent specified herein.

FSC MISC

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SPECIFICATIONS

FEDERAL

PPP-P-291 - Paperboard, Wrapping, Cushioning

MILITARY

MIL-P-116 - Preservation, Methods of
MIL-B-117 - Bags and Sleeves, Interior Packaging
MIL-L-6085 - Lubricating Oil; Instrument, Aircraft, Low Volatility
MIL-O-11773 - Oil, Lubricating, Synthetic (for Impregnating Powder Metal Sleeve Bearings)
MIL-P-14232 - Parts, Equipment and Tools for Ordnance Materiel, Packaging of

STANDARDS

FEDERAL

Fed. Test Method Std. No. 151 - Metals; Test Methods

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Chemical requirements. The brass sinterings, on an oil free basis, shall conform to the chemical requirements shown in table I.

Table I - Chemical composition, percent[1]

Composition	Copper	Lead	Iron	Zinc	Total other elements
1	88.0-91.0	0.05	0.05	Rem.	0.15
3	77.0-80.0	1.0-2.0	0.05	Rem.	0.15
5	68.5-71.5	0.07	0.05	Rem.	0.15

[1]Maximum unless a range is specified.

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3.1.1 Analysis shall be made regularly only for the elements specifically mentioned in table I If, however, the presence of other elements is suspected, or indicated in the course of routine analysis, further analysis shall be made to determine that the total of other elements is not in excess of the limits specified.

3.2 Density. When the density does not vary more than 0.4 grams per cubic centimeter from one section of the sintering to any other section, the overall density shall be within the limits specified in table II.

Table II - Density limits

Class	Composition 1 density-g/cc	Composition 3 density-g/cc	Composition 5 density-g/cc
A	7.3-7.8	7.2-7.7	7.0-7.5
B	7.8 min.	7.7 min.	7.5 min.

3.2.1 Density of critical sections. When the density varies more than 0.4 grams per cubic centimeter from one section of the sintering to any other section., the density requirements of table II shall apply to critical sections where the stresses are highest and not to the overall density of the sintering. The location of the critical section shall be determined by agreement between the contractor and the purchaser.

3.3 Porosity. When specified in the contract or purchase order, the intercommunicating porosity by volume of each sintering shall be not less than 7 percent for class A sinterings. Class B sinterings do not have enough intercommunicating porosity to make it practical to include a porosity requirement.

3.4 Mechanical properties.

3.4.1 Sinterings shall meet such requirements for strength, ductility, impact resistance, hardness and other properties as may be specified in the contract or order or as may be established by tests of sample sinterings which have been demonstrated to be suitable for the intended purpose.

3.4.2 When mechanical requirements are established by tests of sample sinterings, the properties of sinterings furnished under contract shall meet the established requirements within the limits agreed upon between the contractor and the purchaser.

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3.5 Impregnation. When impregnation is specified, the sinterings shall be impregnated with a suitable lubricating oil. When a particular oil such as one conforming to MIL-O-11773 or MIL-L-6085 is specified in the contract or order, the sinterings shall be impregnated with the designated oil.

3.6 Dimensions and tolerances. Sinterings shall conform to the dimensions and tolerances specified on the applicable drawings.

3.7 Workmanship. Sinterings shall be uniform in composition. When cut or fractured, the exposed surface shall conform to standards agreed upon between the purchaser and the contractor. Sinterings shall be free from cracks, laminations, excessive oxidation, and other injurious defects.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Sampling. Samples taken for the purpose of tests prescribed in this specification shall be selected in a manner as to represent correctly the material furnished and avoid needless destruction of finished material when samples representative of the material are available from other sources.

4.2.1 For chemical analysis. At least one two-ounce sample shall be taken from each lot for government analysis. Chips shall be obtained by drilling or milling with clean dry tools, without lubrication. Samples for chemical analysis may be taken after the last sintering operation. When samples for chemical analysis are taken from sinterings after impregnation with oil, either the samples shall be extracted in accordance with 4.4.1.1 or the chips shall be extracted with petroleum ether until all of the oil is removed.

4.2.2 For density and porosity. Five samples from each lot of 10,000 pieces or less, or ten samples from each lot of more than 10,000 pieces shall be tested for density and porosity. Density and porosity shall be determined on the basis of samples free of impregnating compounds.

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4.2.3 For mechanical tests. When mechanical tests are specified in the contract or order, five samples from each lot of 10,000 pieces or less, or ten samples from each lot of more than 10,000 pieces shall be tested.

4.2.4 For visual and dimensional examination. Sampling for visual and dimensional examination shall be in accordance with MIL-STD-105. Inspection level, acceptable quality level (AQL), and classification of defects shall be as specified or approved by the procuring activity. Normal tightened or reduced inspection shall be instituted in accordance with MIL-STD-105.

4.3 Examination. Each sample unit selected in accordance with 4.2.4 shall be examined to determine compliance with the dimensional requirements (see 3.6) and workmanship (see 3.7) of this specification.

4.3.1 Preparation for delivery. Examination and tests for packaging, packing and marking shall be in accordance with section 5 herein.

4.4 Test procedures.

4.4.1 Chemical analysis. The samples selected in accordance with 4.2.1 shall be analyzed by the wet chemical method in accordance with method 111 of Fed. Test Method Std. No. 151 or the spectrochemical method in accordance with method 112 of Fed. Test Method Std. No. 151 to determine conformance with 3.1 In case of dispute, the analysis by the wet method (method 111) shall be the basis for acceptance.

4.4.1.1 Extraction of lubricant. Sinterings to be free of an impregnant for the determination of chemical composition, density, and porosity shall be extracted in a Soxhlet apparatus of suitable size, using toluol or petroleum ether as a solvent. The apparatus shall cycle at least four times after the discoloration by the lubricant of the solvent over the extracted sample has disappeared, but in no case shall extraction be continued for less than 2 hours. After extraction, residual solvent shall be removed by heating the samples in an oven for 1 hour at 250 degrees Fahrenheit plus or minus 10 degrees Fahrenheit.

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4.4.2 Density. The density shall be determined using the following formula:

$$D = \frac{A}{B-C}$$

where D = density in grams per cubic centimeter
A = weight in air of the oil free specimens in grams
B = weight of the oil impregnated specimen in grams
C = weight of the oil impregnated specimen immersed in water in grams

The weights A, B and C shall be determined on an analytical balance to the nearest 0.001 gram or to within 0.10 percent of each weight, whichever is greater. The samples shall be impregnated by immersion for a minimum of 4 hours in oil (viscosity of approximately 200 sec. Saybolt Universal at 100 degrees Fahrenheit) held at a temperature of 180 degrees plus or minus 10 degrees Fahrenheit, and then cooled to room temperature in the oil, or by immersion in oil at room temperature, with the pressure over the oil reduced to an absolute pressure of not more than 2 inches of mercury and held for a minimum of 20 minutes, and the samples left in the oil for at least 10 minutes after atmospheric pressure has been re-established. After impregnation, excess oil shall be removed from the surface of the samples by wiping with a soft cloth. The effect of the surface tension of the water when weighing the sample in water may be minimized by adding a few drops of aerosol to the water.

4.4.3 Porosity. The intercommunication porosity (by volume) shall be determined by the following formula:

$$P = \frac{(B-A) \times 100}{(B-C) \times S}$$

where P = intercommunicating porosity by volume, percent
B = weight of the oil impregnated specimen in grams
A = weight in air of the oil free specimen in grams
C = weight of the oil impregnated specimen immersed in water in grams
S = specific gravity of impregnant at the temperature of the test

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4.4.4 Mechanical.

4.4.4.1 Tension. Unless otherwise specified in the contract or order, all tension tests shall be conducted in accordance with method 211 of Fed. Test Method Std. No. 151.

4.4.4.2 Hardness. Unless otherwise specified in the contract or order, all hardness tests shall be conducted in accordance with method 242 or 243 of Fed. Test Method Std. No. 151.

4.4.4.3 Bend. Unless otherwise specified in the contract or order, all bend tests shall be conducted in accordance with method 231 of Fed. Test Method Std. No. 151.

4.4.4.4 Impact. Unless otherwise specified in the contract or order, all impact tests shall be conducted in accordance with method 221 of Fed. Test Method Std. No. 151.

4.5 Rejection.

4.5.1 Examination defects. Any sample unit having one or more defects shall be rejected. If the number of nonconforming sample units in the sample exceeds the number established in accordance with 4.2.4, the entire lot shall be rejected.

4.5.2 Test failures. A lot shall be rejected for failure to meet any of the test requirements when tested in accordance with 4.4 subject to the provisions of the section on "Rejection and Retests" of Fed. Test Method Std. No. 151.

5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 Levels A and B.

5.1.1.1 Cleaning. Cleaning shall be in accordance with process C-1 of MIL-P-116.

5.1.1.2 Drying. Immediately after cleaning, sinterings shall be thoroughly dried in accordance with any applicable procedure specified in MIL-P-116 that shall not damage the item.

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5.1.1.3 Unit packaging. The quantity of sinterings containing the same stock number to be placed in the unit package shall be specified in the contract or order. Unless otherwise specified in the contract or order, unit packaging shall conform to the requirements specified in MIL-P-14232, except that cushioning material shall conform to PPP-P-291 and bags shall conform to class C of MIL-B-117.

5.1.1.4 Intermediate packaging. The quantity of unit packages, containing items of the same stock number, to be placed in the intermediate package shall be as specified in the contract or order. Unless otherwise specified in the contract or order, intermediate packaging shall be as specified in MIL-P-14232.

5.1.2 Level C.

5.1.2.1 Cleaning and drying. Cleaning and drying shall be as specified in 5.1.1.1 and 5.1.1.2.

5.1.2.2 Packaging. Unless otherwise specified in the contract or order, the packaging shall be as specified in MIL-P-14232.

5.2 Levels A, B and C Packing. Unless otherwise specified in the contract or order, packing for level A, B or C shall conform to the requirements specified in MIL-P-14232.

5.3 Marking. In addition to any special marking required by the contract or order, marking of interior packages and shipping containers shall be in accordance with the requirements specified in MIL-STD-129.

6. NOTES

6.1 Intended use. The sinterings covered by this specification are intended for Army items such as components of time fuzes, fire control instruments and miscellaneous items.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Composition required (see 1.2.1).
(c) Class required (see 1.2.2).
(d) Dimensions (see 3.6).
(e) Whether material is to be packaged and packed for levels A, B or C (see 5.1 and 5.2).

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6.3 The requirements for packaging and packing (see 5.1 and 5.2) and marking for interior packages and shipping containers (see 5.3) specified herein apply to direct shipment for Government activities and apply also, where specified, to contracts or orders between the manufacturer and the Government prime contractor.

6.4 Tests to establish mechanical properties and for acceptance of production sinterings should be determined after consideration of the function which the sintering is to perform and should be made on sample sinterings. Tensile or compressive strength can be determined by placing a suitable load on the sample sintering and determined by deforming the sintering under compression, or by bending, and observing whether fracture occurs before the sintering has exhibited sufficient ductility for the intended purpose. Impact properties may be determined by dropping a weight upon the sintering to be tested. The weight, height of drop, point of impact, and method of supporting the sample should be prescribed, if this method is authorized.

6.5 Sinterings are basically produced by compacting and sintering metal powders. The ultimate properties of the parts largely depend upon the controls and methods used in the processing steps. In the case of copper-zinc materials two sintering methods are available. The older method (hydrogen-sintering) involves the use of a very common lubricant, zinc stearate and sintering in a hydrogen containing gas. The more recent method (nitrogen sintering) involves the use of lithium stearate as a lubricant and sintering in nitrogen. Significant improvements are obtained in mechanical properties by the use of the latter method.

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